

Introduction to Impact Evaluators - CryptoEconDay Paris '22

0:00

what I want to talk about today is impact evaluators which are a specific mechanism to try and get communities of

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agents these could be people organizations or you know various groups

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to collaborate to maximize some measure towards a goal so there's a methyl there

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but the point is how do you coordinate large groups to achieve shared goals and

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impact evaluators provide us a a set of mechanisms to take Community

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oriented funds and allocate them against impact towards uh

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uh measurable uh success towards the goal uh so to set sort of the stage you

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have many uh conditions across lots of problems where you can break down uh the

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problem space into the following model where you have a community of agents that are trying to work together against

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a particular shared objective now they might care about this objective in different distributions so you so each

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uh so you might have like a distribution of of interest in in actually achieving this objective and you have a different

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distribution of capabilities to um affect uh the objective you also have

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a dynamic dynamic environment where the capabilities of the agents change over time so agents might get more capable or

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less capable over time in order to to achieve the objective but what you want

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is to build a system where you can use shared mechanisms to orient the

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community of agents to over time achieve the shared objective uh so what impact developers do is that

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they let you translate a a measure and a shared pool of funds into a pretty

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robust mechanism that can coordinate across many sorts of groups to orient

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and incentivize the the achievement of that objective uh so how are we going to do that and uh

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that sounds like pretty basic but um the uh and impact evaluators are

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basic um but the key Point here is that this very um simple mechanism can cause uh massive

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levels of impact uh and uh to motivate that uh the

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Bitcoin network uh and all of the hash rate and all the consumption of power that the Bitcoin network is causing is

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given by a single impact evaluator which is the uh the block reward of Bitcoin rewarding the hash rate contributed on a

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per block time so um that that's like the level of power that you can get out of like these very

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very very simple mechanisms uh so uh there's impact evaluators are like not well um

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understood as as uh as they haven't like really been studied as mechanisms themselves at least I haven't found um a

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theory of them um there's they have Echoes to to control theory and to other other systems

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um but I've been sort of trying to work on figuring out the the components and and the different like levers that you

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have to tune the impact evaluator so uh this is like the traditional maybe version of it where you have in kind of

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the Bitcoin block reward or the web3 crypto block reward model where you have some reserve of currency you have a

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certain amount of contributions you have some way of evaluating those contributions and you have some way of

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rewarding those contributions in the classic Bitcoin example it would be the reserve is the Bitcoin yet to be minted

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um following some exponential decay so you have some way of like mapping the reserve to

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um to a unit of time the contribute the contributors are all the Bitcoin miners

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that are contributing hash rate uh in a over time you can break it up into rounds and think of contrib taking the

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contributors weighing the impact their impact against against a particular Moment In Time the evaluation function

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in bitcoin's case is super simple and part of why this mechanism I think hasn't been studied well is that it's

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hidden in the in in the model so um in bitcoin's example

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the hash rate matches maps onto the evaluation over

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longer periods of time not in a single block so you don't have an actual evaluation function and a per block time but you have an evaluator that works in

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expectation so as miners compete to Mint blocks on expectation their rewards will

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match up to the power that they've contributed and so you only get this evaluation function working over the

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larger period of time and so you don't the the sort of like impact of other mechanism will sort of um incidental

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the then the reward in bitcoin's case is just uh you the the coinbase minting of

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um the reward to the to the minor address now that model has been adapted to all

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kinds of other blockchains right so um ethereum and many other networks uh follow this model falcon follow this

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model uh in Falcon's case we adapted this model to Mint not just on an exponential decay but take into account

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a um the the reward on a per round basis takes into account a kpi for the network

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which is the actual total storage power that the network is contributing so we're starting to shift the reward

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structure to cause the impact evaluator to be better meaning it's not just going to be an exponential decay that is meant

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to hopefully incentivize that outcome in the long term um mapping the the value of of the

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currency to to the kpi but in addition to that you can measure how well you're doing relative to to some long-term goal

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um and reward based on that so Falcon uses um a kind of this goal against a

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baseline the Baseline here is a certain amount of storage at a particular moment in time and so the reward scales

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relative to how well fog one is advancing against against that that goal

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um so again like you we can redefine the same model in a control theory uh sense

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where you can map to the traditional control theory Loop of having some um let me make that and scale that up

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so everyone can see it

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live slides um so you can have the traditional control theory Loop where you have some

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input signal feeding into a controller the controller is providing some system and some input into some system that's

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going to carry out some action that system is going to Output the action at the same time the system is feeding back

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into into some sensor and the sensor is sending a feedback signal to the controller right so again this is a very

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simple model but it's extremely powerful in terms of the world today operates based on mechanisms this simple like

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many valves like water valves and pressure valves and systems and planes and so on operate

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based on these very very simple structures where you have some way of measuring some State and you can feed

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back that measurement of that state into some actuation to control the system

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so you can think of Impractical uterus as that but for um any arbitrary sets of agents in a in a

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broader Community against some kpi that's measurable again key here is that that kpi is measurable

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another way of thinking about it is as a as a valve so the so far the impact of

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others that I described had a reserve of currency so the Bitcoin example the fact one example and so on all have a certain

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amount of currency that's initialized and then from there um it is uh minted

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out and reduced in ethereum's case you have additional minting in perpetuity but you can you know like plot out some

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you know this moment in time and you can specifically calculate how much eth that's going to mean so think of all of

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those as just having a constant Reserve but you can also think of impact evaluators as having a dynamic Reserve

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where the the actual inflow the the funds of the impact developer has to offer and to align against the incentive

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can change with time so that could change because the impact um achieved is feeding back into the

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evaluator meaning the impact achieved against some goal is producing some Capital return that can feed back into

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the impact evaluator to reinforce the loop if you manage to do that then you can like get these things to like scales

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much faster than Bitcoin um in a sense kind of Bitcoin had this incidental return which is the if

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Bitcoin become appreciates and has more value over time than um certainly people will try to compete

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before it more and you'll get this effect um but it can be much more powerful if you can feed back some amount of the of

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the value creation back into the into the into the evaluator what this would look like is like closing some fraction

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of the outflow over time makes its way back into the inflow and you can kind of um scale these

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um that also suggests that you can form networks of these things uh another view into this like um you

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know kind of the diving into how an impacted value Works uh is that you know there's a mechanism somewhere in in uh

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in a robust way that the community of Agents can trust so for our purposes a blockchain uh can do just fine you have

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a way to deploy a smart contract where community of Agents cannot kind of tamper with a contract and so everyone

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can agree on on doing things um and working against what the contract defines but of course you can create uh

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impact of elevators exist in many other environments you can think of um even like a uh we're talking

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yesterday about you can think of uh conferences in the academic environment as an impact evaluator where like

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there's like a predetermined number of rounds like there's you know one conference a year or two conferences a year uh people submit papers to this

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they get weighed they get sorted then academic credit flows back uh to all the participants that submitted uh papers

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based on you know the the getting inclusion into the into the journal or or Awards like best paper award and so

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on and so and that is like a stable impact evaluator that is recurring year over year over year

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um so well that the community of Agents can bet on that impact of alert continuing to exist um and and there is one of the key

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things here which is this community of Agents is able to maximize that kpi and Achieve that objective

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um if this impact developer is trusted to be around in the long term right so you want a condition where

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agents can individually plan out their future and decide whether or not to contribute towards the uh the

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objective and contribute towards maximizing the measure um independent of all the other agents so the ideally you don't they don't have

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to carry out any other coordination than just uh interacting with the with the impact evaluator themselves

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um so in the uh in the Bitcoin case or the you know the block traditional block reward case you have some long period of

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time you divide it into a set of rounds um you know either you know singular blocks block times or something like that and um and at every round you're

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going to run a pretty simple algorithm which is you're going to measure the world the state of the world and derive kind of you know the the you're going to

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look through and pick out the signals that you care to to reward so you you eval you measure how well the the kpi is

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being maximized then you feed that into a weighing to a evaluate how much

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you're going to reward every participant based on their impact against that kpi and then you pay out those rewards right

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so kind of in a simplistic sense you can boil down like the measure part into

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extracting a kpi and extracting the set of agents that have worked against like API here these are not like single value

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variables with other like distributions right so think of you know P here being a distribution of all the agents in the

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world um way with their you know a measure against the what they contribute to that kpi so the kpi here might be K might be

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like the total hash rate in a particular Moment In Time P might be a distribution of the individual hash rate of each

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agent contributed against that against that case uh then you know you feed that into into

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um uh this kind of evaluation function that weighs that hash rate and maps that

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onto some fraction of the of the funds of the impact of elevator so

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um this might be uh the simplest thing to do is to just uh take the reward

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assign the reward to you know some amount per round and then they distribute that evenly right so you can

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you can think of um sorry I'm getting ahead of myself that's in the reward function um

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uh hearing evaluations just kind of mapping the the impact against the

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um the participants uh I think also here is a hook where you can put in humans in

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the loop to evaluate really complex things so for example there are certain things you can measure uh but often

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translating those measures into a true proper notion of impact requires some

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human Judgment at least a day um and it it can be really useful to feed those

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those measures into a community of Judges who can then um tune the evaluator right so for

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example if you might in a kind of conference Journal case you have a

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standard conference that is happening over time and you have a set of peer reviewers that are evaluating the papers giving their measures of the papers

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doing a very simple evaluation like they have to read every paper and decide for you know strong except we accept uh weak

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reject strong reject um and you have a very simple way to like scale this uh to be able to deal

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with you know some large number of papers and some much smaller number of Judges that are going through and you

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have multiple judges to kind of um get a much better uh sample of like

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what's what's good um and then from there you can aggregate all those scores and decide what what you're going to accept

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um the the more kind of um you could do this for example for things like events right you want to host a crypto econ day

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uh you want to have like many other crypto events around the world you can measure some rewards uh some impact uh

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produced in that event so it could be like the quality of the talks could be the the

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um number of participants that attended this event the ratings that those participants give after an event like

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you know there's a survey you can like take input like that it could be the long-term views on the recordings of of

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this event or something like that um you can but those kpis alone often are very gamable

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um and so you don't want just those measures to to drive a big impact evaluator you want to potentially feed that input into judges who are then

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going to decide the reward distributions and this is where you can aggregate input and then structure it in a way

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where you have limited participation participation of some evaluator judges

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and then of course you like feed that into some reward function um you know you can split up the fund

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you know in terms of rounds this is like the simplest thing to do if you have like some you know number of rounds into the future or you have some other

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function to scale it like exponential decay or or whatever else you want uh and then you kind of decide what each

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participant gets paid out now of course this reward function could could be pretty different right like you can tune

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as we were saying before you can tune the reward function based on the success against the kpi and so on um these kind

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of like simplistic view of like just these parameter spaces it's just to give you an idea of how to construct one of

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these things of course you can feed in you know the state of the world to each one of these functions and um and derive different different

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components um any questions so far

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great sounds pretty basic raise your hand if it sounds pretty basic great this is awesome because that means

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I'm doing a good job because these things are like not well understood yeah like when we introduce this this

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judges concept so this looks like a protocol but then when we introduced this just concept it becomes like a Dao

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kind of structure right yeah yeah I mean the idea here is that you have a protocol that enables

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participation from some evaluator set that is going to be able to put the the

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sensors um since you have these measures right and the the idea the difference here

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between this measure evaluate reward uh picture is that the measure function is extracting some information out of

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the world the evaluate function is Translating that information into impact

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against the kpi and then the reward function is Translating that impact against the kpi into a payout into

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reward payout and so the evaluation part in many cases you need some judgment

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component here that is difficult to automate and so I'm introducing the judges here because there are a lot of

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things that are like not easy to evaluate in a smart contract right so it's difficult to write smart

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contract rules that are going to be so good at capturing the quality of some of some uh progress against the kpi that

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you kind of need judges in the loop to to be able to do that but the idea here is like

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um the point is you can use impact evaluators for extremely complex things because you can have like this very

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intelligent um uh process in between that can weigh extremely difficult to weigh things

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like the conference paper is one is a good example you have like the entire structure of a journal almost automated

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um and you call out to like the the lead people in fields to give their expert

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opinion on what's good in a field and that's a fairly difficult thing to automate because that's a highly Dynamic

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uh thing is the field is constantly changing you require a lot of nuance in terms of picking what's a good

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um a good contribution or not um and oftentimes even today like um you have like panels of Judges that

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that constantly Miss really important things right but um and so things are working well enough but like it's not

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completely automated right it'll be difficult to automate that that process entirely like you can do it with just like the straight up voting uh like if

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you if you um had like I don't know direct direct democracy voting on what's a good paper

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like you would get like a bunch of error output yeah uh so the judge's model is like a

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way to get something better than that now you could you could build all their systems that are much more robust than that

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but it's just just showing it as an example a way of like escaping out to um human judgment

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um any other questions

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uh cool so uh I was mentioning this before like you have potentially constant

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um reserves or constant um uh plots of of Rewards or you can make them variable and the idea with this making them

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variable is that um you could have this might be a way of

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doing long-term public good production when you can only capture the return of

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the public goods many years down the line so you could so for example you could create an impact evaluator against

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um r d on something like filecoin right and you and you want to reward research on cryptographic Primitives Research

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into distributed systems Research into better Hardware

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um whatever you can think of enumerate a ton of things that are important public goods that are going to lead to a better

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filecoin Network and you will only return that value 5 to

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15 years after the initial deployment of reward right or the the initial funding

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of some of some um of some work so you might fund some research you might

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reward some research that is produced but that research has to sift through a big long r d pipeline that is going to

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take many many years to then translate into some product that is going to improve the network and so today you don't have like good

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measures that kind of span this very long period of time however with this you you can do that you can then couple

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some like return of that result back into uh funding back into the impact evaluator it costs more more rewards I'd

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probably be like better examples on this one um in the event case imagine that

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um you're like rewarding the production of like really great events um and then all the events sign up to

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um uh give a fra you know a fraction of proceeds over time generated back to the

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impact evaluator um in in the long in the long term or something like that um

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uh you could also start coupling these impact evaluators into networks so you can think of

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um uh these are like not well lined up because the diagram is not perfect but and you need to like draw more arrows but imagine

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um think think of each of these impact evaluators as a neuron that is doing some processing on a network where

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um you have some measure of some input impact and some reward output and so you

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can start coupling these into more complex networks where the the impact

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out of the the outflows of some impact evaluator are going to get fed into the

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inflows of some other impact evaluator right so if some impact developer is succeeding it might reward other it

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might be rewarded by another impact evaluator so you can start kind of coupling these in interesting ways

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um you can also deal uh so so far as I've described them impact of relators have this problem where

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um they might lead to negative some games so for example if you are going to reward say

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um bringing a lot of new clients uh to The Faculty Network so imagine that you want to incentivize a large sales team

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for the faco network and you create an impact evaluator that rewards um closing some

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individual client you might get into this Zero Sum game where uh people enumerate the potential

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clients and they compete really fiercely for the easiest ones to get um and they they don't spend much of

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attention in the other ones um and that is an inefficiency in the market uh so ideally you want to kind of turn this into a positive sum game where

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you cause this like weak coordination to happen in the environment

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um so so you don't just lead to the traditional Nash equilibrium that happens just out of participants realizing that they're all going to be

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competing against each other and so on you can do something strong like you can shift equilibrium stronger than that by

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um by making the reward super linear so you can make the reward itself scale

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with the impact received so if you instead change that impact evaluator to say

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um at different levels of different clients brought into the network it unlocks different levels of reward you

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then call immediately cause agents to initially enumerate lists of clients sign

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themselves up for different ones and then help each other to to close many more and so you you can cause this uh

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this much stronger positive sum environment there might be a way to like do Pareto

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preferred I.E so not just a positive sum game where participants um or like the net total is is stronger but you can

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actually yield a situation that is prior to Preferred where people are like um enter into like this voluntary

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cooperation spot um and I think a way to do that might be by issuing a rewards based on past

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participation so introducing some memory into the system that can track contributions over time and factor that

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into the into the system um and there might be other ways to do this

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it's just kind of like a speculation uh cool so I think that's oh yeah I had

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um just to kind of examples again um you know kind of the become reward you know hash rate contributor for Block

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time for minor that's like a standard example the Falcon block reward is rewarding the

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it's measuring the QA power contributed per block time and then is rewarding that with some probability

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uh you know you could you could create like this Five Point event reward or something like that

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um great um and that's all I had any questions about IES

24:02

or can everyone does everyone feel like capable of going to like produce some of these things and uh and and uh

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fixing Global problems uh yeah

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hey question for you on the uh the judges piece could you just share a perspective on how well built out that

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space is like is claros an example or is there more work to do in the space of judges for these types of solutions and

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this specifically in in kind of the the blockchain world of like people using judges in in within blockchains I think

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this isn't used enough um I think right now um

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most of these kind of impact related things deployed in blockchains tend to be very simple smart contracts that are

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that are either um trying to speculate about some impact

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um for example like the Bitcoin case it's speculating about the amount of hash rate contributed based on how many

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zeros in a shot 256 function there there are right and so like that's like a very

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clever way to construct an impact evaluator um you but it's coming out of a

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limitation of not thinking about just other ways of measuring um but you and that's a very limited thing right you can't do that for tons

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of other things um so both in the measure function and the value function I think there's all kinds of ways of

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um collecting information that is not at all explored right so another thing you can do here

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um is start I think like imagine being able to start measuring things about the planet uh feeding them into smart

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contracts evaluating some impact reward against that and then and then producing some uh some outputs on a classic

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example here might be you take the landsat imagery you

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um evaluate the amount of forest in various countries and like the the the planet is our

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decisive to reward countries that like increase the area of like Forest right like that's something that very

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straightforward that you could do with with one of these kinds of things um and like that kind of scale output would require feeding the landsat

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imagery running that through object recognition distinguishing like green tarps from 4S

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um you and probably enter into this cat and mouse game where like people get more and more sophisticated trees that

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are plastic to like fool the satellites or the satellite the measure function has to get better

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um and then you uh or you or you can introduce some other model where like you don't just use landsat you use like

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some kind of verifiable claim structure where like you could feed in a instead build a network of verifiers that go out

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and and uh inspect go and inspect right so nation states today use many structures like this to do to do stuff

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like that they have like ranges of inspectors they produce a report they feed that report into some some reward

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function usually the nation state reward function is not positive it's negative right you did something wrong you get a

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negative reward right so so they have inspectors they inspectors measure the world um they evaluate

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um and then you get a negative reward uh but you could you could create positive rewards too right um I guess that the

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other um is not entirely fair to nation states uh there are like strong positive rewards in terms of grant funding like

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you know think of like research funding um for for uh uh long-term r d or

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um carbon credits are a good example where like nation states might buy a whole set of tranches of carbon credits

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um and those are based on kind of um that's less impact evaluatory and more impact uh certificate oriented uh

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but you can think of that those rewards as positive rewards that nation states are using today

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uh yep

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uh when you think about measuring and evaluating those two are very kind of interlinked

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um do you think about first what can I measure and then I create an evaluation

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based on that or do you do you think it's better to think about what should I evaluate and then try and find

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measurements that fit to that evaluation yes your question and um this this the separation of measure evaluate and

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reward is just conceptual for us to kind of be able to separate these things uh like like the already the examples I

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gave around the super linear reward based on the kpi like start putting in the measure into the reward or

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um the uh um judges in a sense will end up doing some amount of measurement themselves as

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well so think of these conceptual definitions as fuzzy uh and and but the key thing is that you need to do some

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kind of sensing of the environment evaluating and relating that to impact um and then causing a reward after the

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fact um I think today like there's a ton of low-hanging fruit you

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can find what's easy to measure you can find what's easy to evaluate and um you

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have blockchain so you have like trivial reward uh very easy to like just create reward structures based on on on

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cryptocurrency right cryptocurrency account payment is like a super simple thing um so like I would just start by saying

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like what do you care about find what's easy to measure and what's easier to evaluate and then you might find things that are both easy to measure and

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evaluate and like that's a great sweet spot to start um but just because something is hard to measure or hard to evaluate doesn't mean

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you shouldn't do it because there might be really valuable impact of others that you might construct right like the Bitcoin example

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um all of that came from was trying to secure the network

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and trying to cause a delay in a Slowdown in a consensus protocol to try and like prevent people from

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just a way of slowing down block production and it was achieved by doing

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this proof of work hash rate thing and it took like 10 years for people like bouncing around those ideas to eventually land on that but the idea of

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coupling the proof of work to a scaled reward on the minting like that's kind of like what what caused the Bitcoin

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success relative to uh B money and all the other things that came before

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um like uh forgot the the charm Digi gold or something like that

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so there were a bunch of examples like that but but you needed like a way of both increasing the number of

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participants in the network and the amount of security that they can treat it and so on you just kind of got out of hand and so now we have this Rogue

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impact of elevator that's like um burning massive amounts of energy and like not doing anything useful with it

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so also be aware like fair warning be careful what you make and evaluate because you are going to get maximization of the function so if your

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function is not very good um you're going to cause like massive networks to to go and and maximize that

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kpi so think of like sets of kpis or building valves that like shut them off or something

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yeah back there

31:05

hello this may be this may drive you off topic and if so I kind of hope it does I

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don't know uh funding the commons and your opening speech last time you talked about like you just said I wish more

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people would pay attention to Dow composability and do stuff and you didn't say anything else about it oh yeah uh how much does this overlap

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conceptually to you I think it does so if it doesn't do you

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want to give me why why it does and then well if dowels are like chat groups with a wallet now when they become something

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more where they're like uh actual apis for them to interface and get real work done between them then the concept of an

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impact evaluator seems to matter like my my straw man example is I create a like

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a customer support Dow that does customer support for web3 projects seems like something you could create an

31:54

evaluator for yeah and and I think especially in in things like customer support or sales or those kinds of

32:02

Industries tend to have already established very good metrics for what success looks like and so that's already

32:09

corporations are using impact evaluators within that structure and so you can just lift that and turn it into a smart

32:14

contract um so as a follow-up can you say 30 seconds or less on what you want from

32:20

dowel composability yes so I think um there's probably a lot here and it's probably thinking about things that I'm

32:25

not remembering in the moment but um I I think right now Dow tooling gives

32:30

you this really useful like you mentioned it's kind of a chat room it's not even a chat room it's like it's a

32:35

set of accounts with a shared treasury right it's like a very simple primitive um and you can plug in different

32:41

functions on the decision making so you can have like direct dial token Wade voting you can have quadratic voting you

32:48

can have many other forms of integrating that thinking you could um you could experiment with much better

32:55

voting structures like or decision making structures where you're distilling out Knowledge from the

33:01

participants and using that knowledge to make a decision which is very different from voting right like voting does not say

33:06

um do what's correct voting says do what people prefer which is not always what's correct and so if you could like distill

33:13

out signals and and couple that to decision making that could be interesting um I think Dao's are

33:20

the Dow tooling represents a missing component that open source projects have needed for a long time which is a way

33:26

for open source to have a shared treasury to of resources to then deploy

33:32

against other things um maybe a way to like deal with impact evaluators like you could you could reward long-term contribution to a

33:38

project right you don't want to reward immediate random contribution because you're going to incentivize noise but you could reward long-term

33:45

contribution as evaluated by the participants in the project or or you might be able to use this

33:51

um that's interesting you might be able to use this as a way to solve this kind of principle agent problem with open

33:57

source projects so um part of the reason why it's very difficult to fund open source is that

34:04

often the community doing the work on the project has a very strong vision of

34:10

where the project should go the funders sometimes like don't know what they want

34:16

um and they have have a gap in in being able to recognize what's corrective to fund and sometimes you get that flipped

34:22

where um funders and stakeholders have a certain set of goals for the software that are like different from

34:29

um from from a set of maintainers and there's a different set of maintainers that like want to take it in a different direction and so you might be able to

34:35

couple these things to orient towards what certain stakeholder communities want out of certain projects right so if

34:41

you wanted to um boot a new operating system and you wanted to get a bunch of device drivers

34:47

to like exist in this operating system you could like um measure the success cases of new

34:53

device drivers being contributed um and and how well they work you need a way of like getting that output and then

34:59

rewarding all the participants that contributed those over time over many years um

35:04

uh so because this gives you a way so if you couple this with the Dow tooling this

35:10

would give you a way of like allowing the measuring and the evaluation piece to factor into the

35:16

software because the it's not just about the individual metrics that you have visible in the in a GitHub repo like

35:22

it's not just the commits it's not just the issues it's not just the comments you want to measure something about the

35:27

the output of the software like how happy are people using this how how much

35:33

value is this creating for for the world in some way and if you have a way of distilling that then you could couple like an impact evaluator with like

35:39

long-term software maintenance um but yeah there's probably a lot of other ways of

35:45

um using Dao tooling to compose with many other many other structures I think like today if there were like much

35:51

simpler websites to construct hours and maybe use the dial tooling without the word dial I think the word Dow is starting to

35:57

scare a lot of people where it just feels like weird and confusing and a lot of the web 2 world just doesn't want to touch Dallas before that reason so just

36:03

call it like a shared wallet and and you know you can open a website and create a shared wallet and you can then couple

36:10

that to all kinds of communities and the function that you decide on how to use

36:16

the wallet or the um the other rights of that like that that shared wallet confers like could be

36:23

totally composable right you don't you don't just want one form of voting you want many ways of like coupling that

36:30

um like the creation of Corporation is an example of like a a shared wallet plus

36:36

um plus some decision making structure around how to deploy the resources

36:42

um and you know like smart contracts dowels are like one or two steps away from creating docs like digital

36:47

autonomous corporations and so on um cool any other thoughts or questions I got one yeah I got the mic uh the

36:56

question is basically like of what type is world and and also and in some context is

37:03

like this very similar idea to the pl Journal Club paper that I mentioned to you last week uh that we talked about

37:09

where world was Blocksworld and it was sort of like a towers of Hanoi type thing you know world and the there's an

37:16

economy of Agents competing to like lift all the blocks basically like solve the game but uh that was a you know 1999

37:22

paper and a you know trivial example so do you have any instincts on like what information would need to be in the

37:28

world strapped or something for this to work yeah in reality like the measure function is it's really kind of closer

37:34

to the system's uh Theory version where like you have a sensor like here the sensor is measuring something about the

37:40

world it's just kind of selecting a subset so um you're not going to measure the entire observable universe but you're

37:46

gonna like you have some you're gonna select the measures that you care about following

37:53

um but but the word world here is just kind of meant to Encompass like it's

37:59

sufficient information to you for you to make that this kind of like evaluation judgment um so whatever you decide to measure is

38:05

what you can evaluate and what you can measure and can evaluate is what you reward so if your reward is not giving

38:12

you good results against your kpi then you probably need to evaluate better or you need to measure better

38:17

um what do you think of a candidate a world like a world where a world is like an ipfs instance in a live P2P instance

38:24

and like the set of things you can measure out of that oh like um wait but but then that's like then you can like

38:30

request anything you want um yeah uh maybe I'm like I think one

38:37

important thing here is that the for this to work in rounds you need to be able to take a snapshot of at a given moment in time so

38:43

um you need to bound the how much information it's like it's kind of like a world snapshot of that moment in time

38:48

um ideally including the information that you care about um or ideally only the information that you care about

38:54

um uh there was a question back there

38:59

sooner than you all right thank you um I was wondering if you have any tips around handling the game theory around

39:07

uh what to measure evaluate and reward because of course just like how rewards

39:12

can be very helpful they could also be very destructive if you get the wrong assumption uh off the bat so yeah so

39:19

um the best tip I have is to iterate like upgrade the mechanism

39:24

um and don't assume that you'll ever get it right so you want you want structures

39:30

that can evolve with time and it'll be very contentious because um

39:36

part of the benefit of creating this impact evaluator is to fix the measurement evaluation and reward enough that the community of Agents can start

39:44

betting about the future and making plans individually betting on on what reward they can get

39:50

so so a concrete example of this is um Falcon storage providers can because

39:57

that block reward is there they can bet on um the the distribution of reward they

40:03

might get and therefore they can bet on facility storage facilities and so on same with Bitcoin miners right they can

40:08

bet on the hash rate being there so they can bet on getting data centers and getting hash

40:15

um uh like Asics and so on and and they can afford to like

40:21

hire a staff of people that are going to operate these machines and so on so that entire like planning of a corporation

40:27

that has to deal with you know many months to many years all of that Horizon

40:32

is embedded within the impact evaluator in a sense and and what you get out of it is that like that constancy of that

40:38

of the thing however like you will run into problems like the Bitcoin case is a

40:43

perfect example um and so you need to make these things upgradable in time to better fit to what

40:49

the community cares about and so I think you need if you find robust ways of um

40:56

uh gleaning kind of the shared preferences of a community and and mapping those into the measures and

41:01

evaluation functions you want to kind of like maybe check along the way how close are those tied and when you notice a way

41:08

like to improve the closeness of those measures then then take that step and improve the evaluator

41:14

um I think other tips like in the meantime because like that's it's hard to write these impact of others with upgradability and whatnot

41:20

um in the meantime I would say like bound the return and bound the amount of money that you give an impact evaluator

41:26

so that like um if it's working really well then you can decide to fund it more and if it's

41:32

not working really well it'll like shut off by running out of money um so like if it's like being bad like

41:38

then you can you like star bit of resources um and that gets closer to the to the intake valve inflow to rad flow model as

41:45

opposed to the you give it a huge Reserve at the beginning um so so this is kind of what I mean by

41:51

you can couple them in networks where um uh you could sort of like evalu the

41:57

wiring here is wrong but um you could measure how well an impact

42:03

developer is functioning and use that to like feed more resources into an impact evaluator

42:09

so basically with having every impact evaluator be

42:14

like I guess able to time out after a while by running out of funds and then only continue the ones that seem

42:20

effective yeah that's like yeah I would apply evolutionary Frameworks to all of these

42:25

things you want to try things see what works observe the environment and then change um thinking like let things die that are

42:32

not good um I have two questions but I'll go quick

42:38

and they're short one are there any tools or Frameworks for simulating the results of incentive structures and kind

42:43

of like modeling it ahead of time before launching um

42:49

not today because the I mean the study of these things is so bad that like this is like the you're looking at the the

42:56

state of the art Theory and impact evaluators which is not very good right like so

43:01

um there are no good like there are totally many good um

43:06

agent system simulation tools that you can use to explore things like this so

43:12

you could take this impact of regular mechanism is very simple so you can probably model these in those systems

43:18

but that's um I don't think anyone's on that work Beyond you know people people simulate these things all the time when they are

43:26

studying some system right so if you're simulating and trying to understand how the Bitcoin network will work or the background Network or whatever you

43:33

implicitly are doing these evaluations but um there is no kind of like current software package that tells you like how

43:39

it you know that is very general and tune to impact of others uh but that's not like a good thing to make

43:44

and then a second question is unrelated to this on a slightly higher level are there any particular problems like kind

43:51

of bigger High scale problems that you think uh not enough people are thinking about or working towards

43:58

sorry um uh that's a very general question and like my mind like exploded with like a

44:04

very long list yeah Unbound complete Unbound uh

44:10

like uh yeah I'm not again like um maybe what I'll say is uh

44:19

right now we're in like this really weird moment in history where we have

44:25

enormous amount of resources super good coordination Tooling in terms of um communication stack

44:31

um very bad coordination tooling when it comes to decision making um preference like finding

44:39

um forecasting um valuing and so on and so we have this

44:45

like whole economy that is rewarding things that are not at all

44:52

well aligned with long-term flourishing of humanity or like living things and so on and so

44:58

today like blockchains and crypto um and Game Theory mechanism design and

45:04

so on um girls economics like the ability to like blend mechanism design with um a monetary system and
and rewards and

45:11

so on like very closely and and do it so in like a world assuming adversarial settings like the crypto part of
crypto

45:17

economics includes like the cryptography world of um groups of agents that are that

45:23

potentially militias or rational and you have to like you get to kind of um build systems like that you can use

45:29

all that tooling to then start getting like bringing closer like the the reward function with with the with a measure of

45:36

like what's good and finding better ways to measure finding better ways of eliciting preferences finding better ways of forecasting finding better ways

45:43

of like rewarding what's good is like really really valuable um like extremely high leverage

45:49

um because you can like if you can close those gaps even like tiny percentages um you know when you multiply it out by

45:55

the the total like production capacity of humanity like this a lot

46:01

um so I don't know it's not very specific oh and probably AI alignment that's like another very concrete one

46:07

that we need more people working on that um um really kind of maybe a little bit of

46:14

another open-ended one is uh around kind of most of the cases where we've seen people trying to do this have resulted in somebody creating some sort of

46:20

inflationary token that's sort of a speculative asset to you know create one of these systems

46:26

um it seems like the inflationary side of that is really valuable to be able to grow a network and get lots of people in and it gives you more control as a

46:32

Creator but at the same time then your reward mechanism which is supposed to stabilize your network is also

46:37

speculative and potentially very volatile any thoughts on how we can yeah that's a good question um so

46:43

another thing that's going on in in in bitcoin's case because it's not just an impact evaluator it's an effective

46:49

relator plus the resource that's being contributed is expensive

46:54

um so in these kind of traditional block reward models um you have a

46:59

like you you have Capital inflows transitioning as in like inflows into

47:04

the network through through them through that impact evaluator so through the block reward

47:10

um and so they they they're a way of mapping the this sort of like external

47:15

value into into the network and into the currency without having to go through like just speculative changes or things

47:22

like that um I think part of the part of that speculation is probably

47:28

um like Bitcoin for example like the the you have this

47:33

the speculation is what's causing the hash rate to increase and then the hash rate increase is then giving some shock

47:39

absorption to the to the economy and so the more speculation there is the more

47:45

hash rate will appear and then the more there will be a shock observer on on that economy

47:50

um the particular Bitcoin one because it busts really bad like the a Bitcoin bust is really damaging like if

47:57

a lot of Bitcoin miners are um um failing out then it takes a while to

48:03

like bring them back into the network so so um I think I I can't confirm this

48:08

because I don't have enough information but um I think that's part of what's going on with the Bitcoin winters like um the

48:14

hacking rate and Bitcoin miners busing um and the recovery period of that entire

48:20

network I think is what's dragging out the the Bitcoin uh Gap and if you had a

48:26

much smoother and it's part of the reason why we have like all kinds of systems and filecoins are like

48:32

um dampen and spread out those effects like for example the the block reward is like a smooth exponential decay not like

48:38

this having like this crazy having rate um every four years which is like you know like you do not design economic

48:44

system to that in mind unless you want a lot of volatility um so that kind of dampening

48:50

um or collateral structures that spread out the the the

48:56

um the impact that can produce like significant absorption to that and then that can stabilize even better

49:01

um so I think things like that are useful I mean beyond that like there's a separate question which is so those are

49:06

like systems within the the crypto economics separate from all of that like create utility like do something that

49:12

where the impact that you're measuring is valuable right so in the in the 51 example the impact that we're measuring is capacity contributed to the network

49:19

and now with lotcoin plus um actual deployment towards storing valuable data for long periods of time

49:26

right so and that's what the impact evaluator of the block reward and popcorn is measuring it's measuring like

49:31

bringing in a lot of capacity to the network that you can use it's not just wasted work it's capacity you can use

49:37

it also brings security to the network um and it's saying great like also use

49:42

that capacity to store really valuable data that's like really good impact for what 511s would be and so that's an

49:48

impact evaluator that's much more closely coupled with what the network wants to do um yeah I know that that's a good example

49:55

but yeah you want to keep going or yeah we can keep going is there any more questions I don't have all night but

50:04

all right yeah over here

50:09

suppose like if we if there's a reward function so how do we solve this paper clip optimization problem in this case

50:16

oh yeah this is like a this is how you make a paperclip optimizers like be

50:21

careful um that I mean judges

50:26

uh or like building the traps that we were discussing before like the switch switches like you let it run out of money or you building some safety

50:33

switches and so on blockchains in general should have safety switches you you have like the really nice

50:39

um model that at the end of the day you can like everyone can just like agree to not run it anymore which is like you

50:44

need some other system to coordinate that kind of action um but you can do that so uh like August

50:50

models uh which I think um if enough participants in the network said that a that a prediction

50:57

um that a prediction uh bet wasn't ethical would like shut it off was a very good example of a good mechanism to

51:03

to ensure safety um like it's not very sophisticated but it like works pretty well right like auger

51:09

has not had um at least my knowledge hasn't had very significant like backfiring of of that because there's this inherent problem in

51:15

prediction markets where like the moment you can um take any predictive bet you can cause all kinds of other bad outcomes out of

51:22

it you can like predict what a bad outcome put a lot of money against the opposite bet and then cause it to happen

51:27

and so um that unethical switch in auger helped prevent that

51:34

problem but uh I mean more broadly like

51:40

um avoiding paper clip optimizers really means because of what I'm like the

51:45

paperclip Optimizer problem says you have an AGI system that is like

51:51

controlled by a very simple agent Loop that's just maximizing a very simple function and so that inherent problem

51:57

like in order to solve it you have to flip it you have to like say the the the

52:02

intelligent has to define the optimization function and so that's like giving the intelligence function the

52:09

ability to rewrite its own evaluation function and like that's also scary for other reasons which is then you have

52:16

like zero control um so I don't know it's kind of like anybody I don't know if it's an impossibility result but it's it's like

52:22

you either have a measure that's really simple that's like not going to be correct or

52:27

you let the system derive its own measure um that and like just

52:32

who knows uh hopefully it'll figure out what's good um

52:38

but it like reduces to like is can you do you get ethics out of

52:45

understanding the world and there's some this is the known as like the orthogonality thesis in um in

52:52

Boston's work which is like dividing um in other people's work which is like dividing the like your knowledge like

52:58

ideally you want the knowledge of the world to increase like the ethical actions so like the

53:04

more knowledge you have of the state of the universe the more ethical your actions become um the problem and so the orthogonality

53:12

thesis says that there is no so no correlation there um but that's a thesis and it's not improved and I think like a lot of our

53:19

future rests on whether or not that's true so fun times Yep this is taking a very weird

53:25

Direction yes um so with the judges part of the

53:31

evaluate function um there's the human element and judgment may not always be objective

53:39

um it could be subjective and like different between different people so how

53:45

do we how does the impact evaluate or not break yeah I mean I mean all measures and all evaluations will be

53:52

lossy and we'll have some information like um you'll create some Gap ideally the error rate is very low so

53:59

um a way to solve that is like by building better better rubrics for judges this reduces to like all the ways

54:05

in which judging oriented systems have tried to do this which is like you build

54:10

um good rubrics for what it means to be a good judge you build ways of replacing judges you build ways of like having a

54:17

larger number of Judges that capture more of the space um and and so on right so

54:25

um like the judge here is a escape hatch that lets you feed in a lot

54:31

more complex evaluation work into a very simple mechanism um

54:36

with the hope of like that that is like already better than whatever you were going to be able to write into this

54:42

function right um judges can very likely be better off than like a very simple thing like if

54:48

you create an impact evaluator and just say it's just going to reward the number of events that you host then you're

54:54

going to immediately get a lot of events of like one person hosting thousands of events per day right and like that's not

55:00

uh that's not quite capturing what you meant so you're like okay well at least a certain number of people have to attend but then just measuring that then

55:07

it's like okay well you're gonna get a lot of events that just have a lot of people coming and going really quickly but they're not actually creating any

55:13

value um and so you you keep like having to to evolve your notion of what you're measuring and what you're evaluating

55:20

um and judges are like this like kind of like um uh like uh

55:27

ability like the skip hatch that lets you say for whatever metrics you have feed them to like a very Intel like a

55:34

group of very intelligent evaluators that that can like figure out what the preference of the

55:40

group are and you can use and you can evolve the rubrics in a in a way um but you know not perfect by any means

55:45

I'm not objective like not only are they going to be subject intersubjective um they won't even necessarily be

55:51

aligned with the goal set of the community I mean there's a whole separate part of this which is

55:56

we don't have good Solutions yet for how to take very large communities of agents

56:01

and get them to agree on preferences like they tend to have like the the larger the community you make the more

56:07

they disagree about preference sets um today um

56:12

I had a question so um as I see this is a part of the file coin

56:18

talk I was wondering um how can you see um what metrics in the falcoin network

56:23

would be good to measure evaluate and reward and who would that benefit

56:30

yeah I mean um there's all kinds of things I mean the very basic one that's not going to work measures any values and rewards is the bringing of lots of

56:37

storage both capacity and actual usage uh into the network there's all kinds of other things like retrieving um serving

56:43

data really quickly um the you could evaluate like applications and application development

56:49

you have to weigh but again you don't just want number of applications you want to weigh that by some measure of

56:55

utility of each application um and uh you could reward the site the the

57:02

size of the community and the and the the the the the um

57:07

the breadth of the community around the world you could reward

57:12

um you could reward r d in the long term you could reward like the the building of the network itself in a very long

57:19

term um uh times you could um yeah a lot of a lot of these impact

57:25

developers within a network are going to be mapped onto what's good for that Network um and so I would like you could

57:33

broaden your definition of a network and say could you create an impact evaluator on falcoin that rewards something outside of filecoin like and the answer

57:40

is yes and we're doing it already with something like five point green so that's an example where you're saying hey this is like really bad problem on

57:48

the planet in the world um we're trying to like and and our system is going to cause

57:54

some impact on that world can we both like like undo any damage that we might cause

57:59

and or ideally cause zero damage or can we actually go further and cause a good

58:04

impact against that measure so can you like actually start like decarbonizing the planet by having more fat coins so

58:10

like by by with Five Points like ideally correlate the success of outgoing with like decarbonization right and if you

58:16

can do that then then you're now having impact outside of the network into a much larger picture and you can

58:21

definitely do this um but you're now like vaulting you're using the Community Resources to impact

58:27

this broader problem and so that's something that the community of Agents has to agree on it just so happens that in Falcon's case popcorn is not embedded

58:33

like it is embedded in like a planet where all the community of agents are like gonna suffer if like this problem

58:39

isn't fixed so like you could argue it's like fundamental to the continuation of falcoin that like the agents continue

58:45

but it's kind of like saying um you know like modern corporations and

58:50

capital should like all invest in AI reduction of AI risk because that's a

58:56

way of preserving all shareholder existence right and it's like that's not going to fly and they'll record um

59:03

although it's so much to try that that'd be interesting uh

59:09

certainly that's what alphabet should be doing but you know so it's interesting that we had a

59:14

question about subjectivity and then AI because my question combines both of those topics um so I was having a

59:21

conversation with someone earlier and we were talking about evaluation and I was arguing how I believe that human

59:26

evaluators are optimal to have in a circumstance like this I mean evaluating pretty much everything because I don't

59:33

know goodness is subjective it's it's hard to have computers to make decisions like what is good what is bad but they

59:39

were arguing that they think that with the current state of ai ai would be you

59:45

know less subjective more objective about uh you know what is good and

59:51

should be rewarded versus what is bad as long as you feed it the right parameters around what your goals are so I was just

59:58

wondering like with your experience if you agree with the you know benefit of

1:00:03

having AI evaluators versus human evaluators um so so the I give like a

1:00:10

I won't give a long answer because I would require like writing books um but I will give like a short and maybe like

1:00:16

a slightly short less short answer so the very short answer is um uh humans and digital computers are

1:00:25

um like you you can have intelligent systems that gather information and make

1:00:31

some decision against like a goal um but all kinds of things can go wrong and they have different currently

1:00:37

there's different failure modes so um like uh narrow AI so like non-agi AI

1:00:43

systems um what they're really good at is processing a set of data and then

1:00:49

optimizing against a function that you're giving and then repeatably continuing to apply that same function

1:00:56

right so most digital systems that we've made are very basic in terms of following instructions very concretely

1:01:03

so if you can there's like this programmer joke where like if a computer causes an error it's not because the

1:01:10

computer decided to do something wrong it's like you told it to do the wrong thing and so it's your job to like figure out how to tell it to do the

1:01:15

right thing so that it does it um and so in a sense it's like how well those AI models are going to

1:01:21

evaluate something reduces to how well the optimization function captures what

1:01:27

you really want to want to evaluate and then humans have this other benefit and

1:01:33

problem so the benefit is they have a much broader context of what's good and so even if you give them a bad objective

1:01:39

a bad function to optimize like if you give judges a rubric that's pretty bad the judges will start deviating from the

1:01:46

rubric and start optimizing for what they think is good and sometimes that's good and sometimes that's bad sometimes

1:01:51

it's really good because you have like um thoughtful uh well-intentioned judges that are going to produce a better

1:01:57

outcome sometimes you have bad judges that are going to like decide really like things that are totally

1:02:03

uncorrelated with what you wanted them to do and it's just more correlated with like their interests or some other

1:02:08

interests um and so when you have that like that outcome um humans give you like the full

1:02:14

generality of like anything that a human can do which is like a double-edged sword for sure and so you have you can

1:02:21

lead to a much more comprehensive measure of what the the whole system is able to do because like that evaluate

1:02:27

function suddenly goes like hold on like what is this impact evaluator doing is this good like should we be even doing

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this like all that is available within the that function um and so that's like a super powerful

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choice um but it also falls into all kind of

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other biases that will creep in and or not even just biases like you you could

1:02:49

have like rational or relation sectors that are now working against the goal of the system so um in that case like a

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very narrow AI system will like perform more more predictably not necessarily better it's just like more predictable

1:03:02

um so it's kind of like a new like a I said short that's that's like that was like the last short answer the the longer view is like

1:03:09

um in the long term um but uh humans are still like

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processing information and making decisions in the same way that computers process information and make decisions

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and so a lot of it go a lot of the distinction is just in how we're measuring how we're processing and as

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the capabilities of AI increase like um artificial intelligence uh is like

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guaranteed or almost guaranteed to um Encompass everything that Humanity can do and like that Gap just means like

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anything that a human can do on AGI can do and so there's like the dumb definition which is like oh if you think

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humans can be good at this or bad at this and AI like some AI system can do it it's just not the AI systems that are

1:03:53

getting deployed today though the i o systems are getting deployed today are like really narrow models thank you